Hritish Mahajan

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EDUCATION

University of Illinois Urbana-Champaign **Aug 2026 - Dec 2027**
Master of Computer Science *Urbana, IL*

- **Coursework:** Applied Machine Learning, Deep Learning for Healthcare, Database Systems

Rochester Institute of Technology **Aug 2021 - May 2025**
Bachelor of Science in Computing & Information Technologies *Rochester, NY*

- **GPA:** 3.80/4.00, Summa Cum Laude, Dean's List All Semesters
- **Leadership:** Tech Director - RIT Student Government | Tech Lead - GDG @ RIT | Vice President - RIT Chess Club

EXPERIENCE

AI Engineer **Feb 2026 - Jul 2026**
Deriv Dubai, UAE

- Built and shipped two **company-wide AI platforms** on **GCP** for the Finance team, handling **3,000+** requests and invoices/month across **$3M+** in volume.
- Worked closely with the **Procurement** and **AP** teams to automate their full lifecycle, from vendor sourcing and outreach to PO tracking, invoice validation & payment tracking, cutting processing time by **40%** and saving **$150K+/year**.
- Automated approvals, vendor risk checks, and invoice **OCR** with **Gemini** and **Vertex AI**, cutting inference costs by **~60%**.

Software Engineer (Part-time) **Dec 2025 - Apr 2026**
Virufy San Francisco, CA (Remote)

- Built and shipped a clinician-facing **React** app **deployed in 2 hospitals** for audio collection and patient annotation, handling PHI/PII for **5,000+** cough, voice, and breath samples for respiratory disease screening.
- Designed a serverless **AWS** ingestion pipeline (**Lambda, S3, DynamoDB, SageMaker, Cognito**) feeding multi-hospital data into **TensorFlow** models for pneumonia, asthma, and COVID-19 detection.

Machine Learning Engineer (Contract) **Jun 2025 - Nov 2025**
Circum Life Sciences Nice, France (Remote)

- Architected the full software stack for **Predisens**, a ML-powered sepsis detection device (IV catheter + wearable patch) for ICUs.
- Trained early-sepsis prediction models on **200k+** ICU patient records (MIMIC-IV, eICU), reaching **AUROC ~0.89** for Sepsis-3 onset within 24 hrs.
- Built a real-time inference pipeline (**FastAPI**, **PyTorch**) streaming ESP32 biosensor data to a **React Native** app with **<200ms** latency, designed for **HIPAA** and **ISO 13485** compliance.

Software Engineer Intern **Jun 2024 - Aug 2024**
Global Payments Inc. Atlanta, GA

- Automated utility charge calculations across **10K+** units using **C#**, **.NET**, PHP, and SQL Server, replacing manual billing work for property managers.
- Refactored billing and utility management workflows and shipped changes through **Jenkins** CI/CD pipelines.
- Led a team of **3 interns** to build a mobile app driving community engagement among Gen Z residents.

Software Engineer Intern **May 2023 - Aug 2023**
Unity Infotech Dubai, UAE

- Cut app load time by **~40%** by porting a mobile app to **Flutter Web** and refactoring client and UI logic.
- Built account management features for a corporate banking app using **Flutter**, **.NET Core**, and SQL Server.
- Implemented geotagging and live delivery routing for a grocery app using **Google Cloud** APIs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, Java, Dart, Bash
Frameworks: React, React Native, Next.js, Node.js, FastAPI, Flask, .NET, Flutter
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, HuggingFace, pandas, NumPy

LLMs & Agents: RAG, Tool Calling, LangChain, MCP, Agent Workflows, OpenClaw, Claude Code, Cursor, LoRA Fine-tuning, Vector DBs
Cloud & DevOps: Docker, GCP (Vertex AI, Cloud SQL), AWS, PostgreSQL, MongoDB, CI/CD